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CLINICAL REVIEW

# Interplay between social media use, sleep quality, and mental health in youth: A systematic review

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SUMMARY

Social media applications are increasingly prominent among youth. This systematic review provides a comprehensive assessment of the literature on the relationship between active social media use, sleep quality, and common mental health outcomes (anxiety, depression, and psychological distress) among youth. MEDLINE, PsychINFO, EMBASE and Scopus were searched for observational studies investigating this relationship among youth (aged 16–25). Thirty-six cross-sectional studies and six prospective cohort studies met the inclusion criteria. Among cross-sectional studies, significant associations between excessive social media use with poor mental health outcomes (n = 33), poor sleep quality (n = 24), and significant associations between poor sleep quality and negative mental health (n = 16) were found. In longitudinal studies, frequent social media use was a risk factor for both poor mental health (n = 6) and poor sleep outcomes (n = 5). Some studies showed sleep quality mediating the relationship between social media use and negative mental health outcomes in youth. Overall, included evidence links excessive social media use to poor sleep quality and negative mental health in youth. Given the public health implications of sleep problems, excessive social media use warrants further investigation to clarify the directionality and strength of their associations with poor sleep quality and negative mental health outcomes.

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## Introduction

Restful sleep is necessary for optimal physical and mental health. However, insufficient sleep is a prevalent and pressing health issue in today's fast-paced connected society [1]. Sleep loss generally refers to sleep duration shorter than the seven to 9 h necessary for restful sleep, whereas sleep disturbance refers to poor sleep quality that hinders daily functioning [2]. Poor sleep is associated with excessive daytime sleepiness, depressed mood, and difficulty concentrating [3], as well as more insidious, long-term outcomes such as chronic disease [2]. In Canada, 32% of adults (aged 18–64) reported sleeping for fewer than 7 h per night [4] and approximately 70 million adults in the US and around 45 million in Europe are suffering from a chronic sleep disorder [1,5–7]. This is a particularly important issue for youth, as sleep disruption has a negative effect on psychosocial health and may promote risky behaviors such as abuse of nicotine and marijuana [8]. However, the current literature suggests links between sleep hygiene and psychological issues to be bidirectional [9]. Thus, the relationship between poor sleep and emotional and behavioral difficulties in youth should be further investigated.

Recently, there has been a growing body of literature surrounding social media use and its effects on sleep quality and mental health in youth. Social media are defined as Internet-based applications that facilitate exchange and exploration of user-generated content [10,11]. Social media and other internet-based technologies are a pervasive part of today's connected world with the greatest adopters being young adults, 60% of whom report using screens within an hour before bedtime [12,13]. According to a recent report, 81% of youth report use social media [14]. By virtue of being interactive, social media platforms have their benefits (e.g., spreading and engaging in healthy practices) and risks

*Abbreviations:* ADHD, attention deficit hyperactivity disorder; CLARITY, Clinical Advances Through Research and Information Translation; FOMO, fear of missing out; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

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(e.g., sedentary behavior, longer sleep latency, depression) [15]. Adolescence is characterized by a need for identity development and peer acceptance, which social media currently plays a role in facilitating, often at the expense of sleep and sound mental health [15,16].

The current body of evidence on the directionality and relationships between social media use, mental health, and sleep is inconclusive. It has been suggested that the amount of time spent online can have bidirectional effects on depressive symptoms and attention deficit hyperactivity disorder (ADHD), and this risk is particularly heightened in those with pre-existing poor mental health [17]. Moreover, it has been suggested that social media use closer to bedtime is associated with increased cognitive arousal, leading to delayed sleep latency and difficulties maintaining sleep [18]. Several hypotheses have been posed regarding this relationship, including time spent online "displacing" time spent on sleep and face-to-face interactions, compensating for lack of social skills and inability to sleep, and an obligation to always be available [18,19]. Youth are most prone to peer pressure as they are still developing a sense of self-regulation as they mature. As such, they may be at higher risk for adverse effects of social media use, notably poorer sleep and mental health outcomes [20,21]. Therefore, knowing the effects of social media use on sleep and mental health is important for both informing health policies and personal decisions, as well as assessing social media as a facilitator for health improvement.

The objective of this systematic review was to describe the association between social media use, sleep quality, and mental health in youth. As a secondary objective, we also sought evidence on the role of these variables as mediators, moderators, or confounders of this relationship.

## Methods

### *Protocol and registration*

The protocol for this review was registered with the International Prospective Register of Systematic Reviews (Prospero; CRD42019127527) on March 5, 2019. This review follows the reporting guidelines outlined by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) checklist [22].

### *Search strategy*

We searched MEDLINE, PsychInfo, EMBASE, and Scopus electronic databases in February 2019 using a combination of controlled vocabulary and keywords, and the search was updated in November 2019. Forwards and backwards citation tracing for included studies was also conducted to identify relevant studies missed by the electronic search strategy, lasting until April 2020. The full search strategy for each database is provided in the online supplement ("See Supplementary Table S1").

### *Inclusion and exclusion criteria*

We included studies that assessed active social media use, sleep quality, and common mental health conditions, including depression, anxiety, and psychological distress. All three variables must have been reported, with variables for sleep and mental health measured in each study using validated scales (e.g., Pittsburgh Sleep Quality Index, the Beck Depression Inventory, etc.). Although our age group of interest was youth aged 16–25 years, we included studies that used an age range of 12–30 years to ensure we did not miss any relevant information. Finally, we included observational studies that were published after 1990, which was the year when many social media sites were established [23,24]. We did not impose any restrictions on the language of publication.

We excluded studies if they primarily focused on the use of passive social media (e.g., watching TV, listening to music, etc) and electronic devices not capable of accessing social media (e.g., video games). We also excluded studies assessing clinical sleep disorders (e.g., insomnia, restless legs syndrome, etc.) and those focused on other mental disorders (e.g., psychotic disorders, eating disorders). Studies which were not peer-reviewed and dissertations were not included in our review.

### *Study selection*

Two reviewers (RA & JH) independently retrieved citations and screened titles for inclusion. All selected papers from the title screen were saved on Mendeley reference management software, where both reviewers independently screened and reviewed abstracts using a standard form. Disagreements on selected abstracts were resolved through discussion and consensus between the two reviewers.

### *Data extraction and risk of bias assessment*

Data extraction for eligible studies was conducted independently by one reviewer (RA) using a standard form on Microsoft Excel and was verified by a second reviewer (JH). Data on study authors, publication date, study design (cross-sectional or prospective cohort study), location of study, sample size, source of sample (nationally representative or convenience sample, etc.), the measurement tools employed, and demographic information of the study population were extracted from full-text articles. The main findings – including activity and duration of social media use, sleep quality and duration scores, as well as anxiety, depression and psychological distress scores – were also extracted. Discrepancies in the extracted data were resolved through discussion with independent reviewers (KA & SS).

The two reviewers (RA & JH) independently conducted risk of bias assessment for all included studies using CLARITY (Clinical Advances Through Research and Information Translation) tools [25,26]. For each study, the CLARITY assessment tools gave a rating of low ("Definitely Yes"), moderate ("Probably Yes" or "Probably No"), or high ("Definitely No") risk of bias for each domain. Any discrepancies between the independent risk of bias assessments was resolved through discussion and consensus.

### *Data synthesis and analysis*

We synthesized the data through narrative discussion and descriptive tables on the associations between social media use, sleep quality, and mental health. We did not conduct a meta-analysis of the findings as measures of association and tools used to assess social media, sleep quality, and mental health problems varied widely among included studies, making it difficult to statistically synthesize the results [27].

## Results

### *Study selection*

After removing duplicates, 1176 articles were identified from the four databases. Forty-three articles were selected for full-text screening, of which 33 met our inclusion criteria. An updated search was also conducted in which we identified one relevant article. Eight additional studies were identified through forwards and backwards citation tracing for a total of 42 included studies. An

overview of our study selection process, along with reasons for exclusion, is outlined in the PRISMA flowchart (Fig. 1).

*Study characteristics*

All studies examined relationships between social media use, sleep and mental health ("See Table 1"). The paper by Hokby et al. was included as both a cross-sectional and cohort study [28]. Few studies (n = 5/42) used nationally representative samples, with the majority (n = 29/42) using samples from a specific region or location. Ages of participants ranged from 11 to 29 years old and most studies (n = 27/42) included approximately an equal proportion of males and females. All studies assessed symptoms of anxiety, depression and psychological distress. Self-reported sleep quality (n = 17/42) and sleep duration, including weekday and weekend bedtime (n = 17/42) were most commonly measured. Some studies (n = 10/42) also assessed sleep disturbances, including sleep fragmentation, difficulty initiating sleep and sleepiness during the day.

Most cohort studies (n = 4/6) focused on symptoms of anxiety and depression, with two also examining symptoms of psychological distress [28,29]. Social media measures were mostly self-reported or assessed using questionnaires and one study [30] used the Young Internet Addiction test. Sleep and mental health were assessed via validated scales or self-reported questionnaires. Among cross-sectional studies, fifteen investigated internet use [24,28,31–43], fourteen examined mobile phone use [44–57], and five assessed general social media use [58–62]. Mamun and

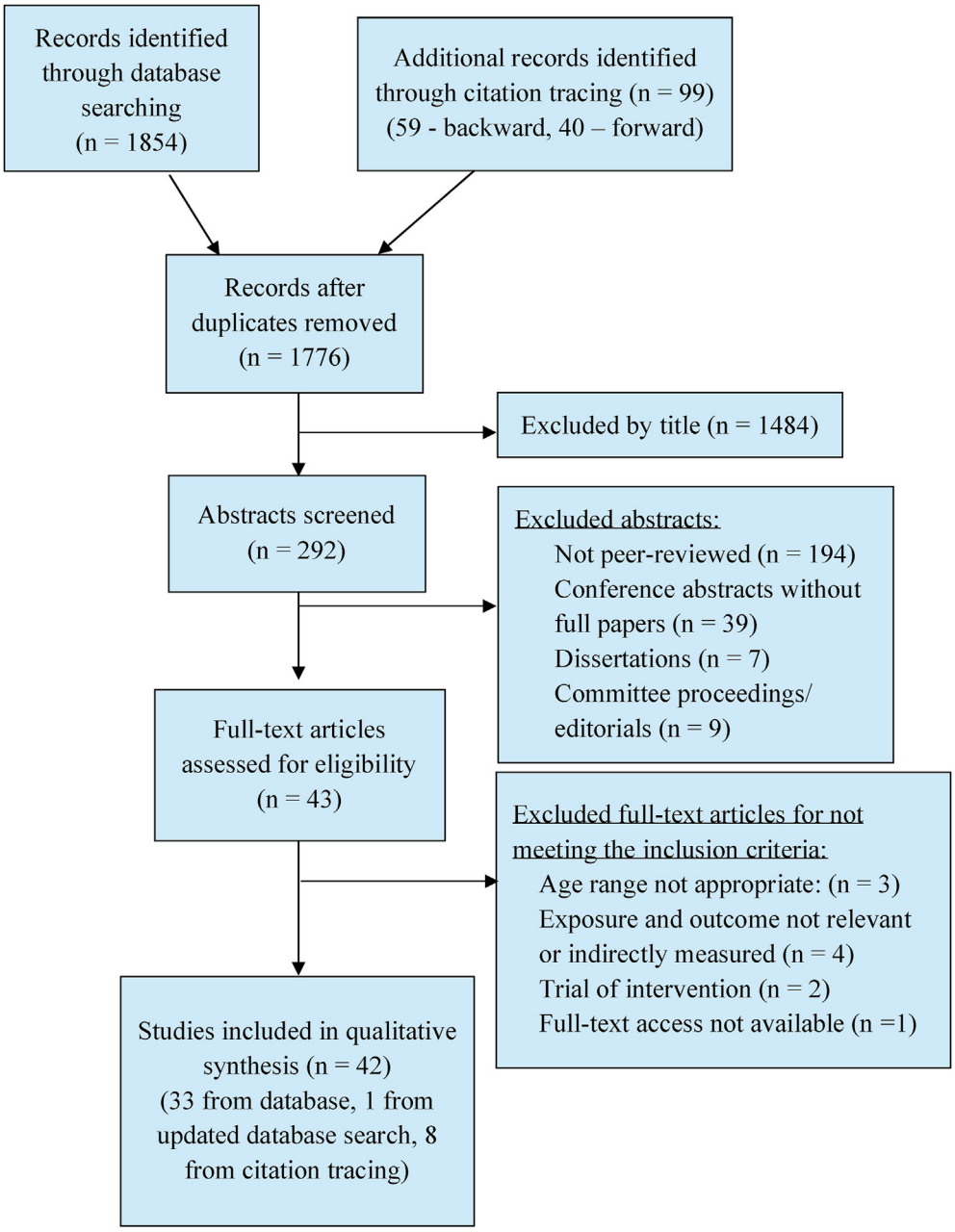



**Fig. 1.** PRISMA flowchart outlining study selection.

**Table 1**
Characteristics of eligible studies.

| Author | Country/Region | Study Design | Sample size | Sample Source | Sample Characteristics | |
|---|---|---|---|---|---|---|
| | | | | | Age (Mean or Range) | % M |
| Adams and Kisler, 2013 [45] | USA | C–S | 236 | Convenience sample | Mean age (yr): 22.1 (SD: 4.2) | 20% |
| Belanger et al., 2011 [31] | Switzerland | PC | 7122 | 2002 Swiss Multicenter Adolescent Survey on Health database[c] | Age (yr): 16-20 | 54% |
| Bhandari et al., 2017 [37] | Nepal | C–S | 937 | Universities | Mean age (yr): 21.0 (SD: 2.2) | 45% |
| Brunborg et al., 2011 [63] | Norway | C–S | 816 | Norwegian National Register[c] | Mean age (yr): 27.9 (SD: 7.4) | 44% |
| Chen et al., 2017 [55] | China | C–S | 1441 | College | Mean age (yr): 19.7 (SD: 1.4) | 48% |
| Cheung and Wong, 2011 [33] | Hong Kong | C–S | 719 | High school | Mean age (yr): 14.1 (SD: 2.0) | 60% |
| Demirci et al., 2015 [44] | Turkey | C–S | 319 | University | Mean age (yr): 20.5 (SD: 2.5) | 36% |
| Dewi et al., 2018 [56] | Indonesia | C–S | 714 | Universities | Mean age (yr): 16.0 | 43% |
| Do et al., 2013 [34] | South Korea | C–S | 136,589 | 2008–2009 Korea Youth Behavioural Risk Factor Survey[c] | Age (yr): 13-18 | 52% |
| Eyvazlou et al., 2016 [57] | Iran | C–S | 450 | Universities | Mean age (yr): 20.4 (SD: 1.6) | 36% |
| Hokby et al., 2016 [28] | Estonia, Hungary, Italy, Lithuanian, Spain, Sweden, UK | C–S, PC | 1554 | State schools | Mean age (yr): 15.9 (SD: 0.9) | 44% |
| Ikeda and Nakamura, 2014 [52] | Japan | C–S | 2698 | High school | Age (yr): ~15-18 | – |
| Islam and Hossin, 2016 [42] | Bangladesh | CS | 573 | University | Age (yr): 20-30 | 70% |
| Kadam et al., 2016 [58] | India | CS | 900 | College | Mean age (yr): 19.3 (SD: 1.5) | 43% |
| Kitazawa et al., 2018 [35] | Japan | C–S | 1258 | Universities | Mean age (yr): 19.3 (SD: 1.1) | 43% |
| Kojima et al., 2019 [39] | Japan | C–S | 2887 | High schools | Aged (yr): ~12-15 | 50% |
| Kootesh et al., 2016 [38] | Iran | C–S | 205 | University | Age (yr): ~22-27 | 48% |
| Lam and Peng, 2010 [30] | China | PC | 1041 | High school | Age (yr): 13-18 | 47% |
| Lange et al., 2017 [59] | Germany | C–S | 7533 | The German Health Interview & Examination Survey for Children & Adolescents Study[c] | Mean age (yr): 14.2 (SE: 0.01) | 51% |
| Lemola et al., 2014 [60] | Switzerland | C–S | 362 | High school | Mean age (yr): 14.8 (SD: 1.3) | 55% |
| Li et al., 2017 [40] | China | C–S | 1015 | Convenience sample | Age (yr): ~12-17 | 59% |
| Liu et al., 2019 [69] | China | PC | 3396 | College | Mean age (yr): 18.3 (SD: 1.7) | 81% |
| Mamun and Griffiths, 2019 [64] | Bangladesh | C–S | 300 | Convenience sample | Mean age (yr): 20.7 (SD:1.7) | 61% |
| Mamun et al., 2019 [43] | Bangladesh | C–S | 405 | University | Mean age (yr): 20.2 (SD: 1.6) | 50% |
| Nishida et al., 2019 [51] | Japan | C–S | 295 | High school | Mean age (yr): 16.2 (SD: 0.9) | 57% |
| Park et al., 2018 [36] | South Korea | C–S | 766 | High school & middle school | Mean age (yr): 15.1 (SD: 1.4) | 63% |
| Peltzer and Pengpid, 2015 [32] | Africa, Asia, Caribbean, Latin America | C–S | 20,222 | Universities | Mean age (yr): 20.8 (SD: 1.6) | 42% |
| Rod et al., 2018 [67] | Denmark | PC | 815 | University | Mean age (yr): 21.6 | 68% |
| Rosen et al., 2016 [46] | USA | C–S | 310 | University | Mean age (yr): 25.9 (SD: 6.6) | 42% |
| Seo et al., 2017 [61] | South Korea | C–S | 24,676 | High School[c] | Mean age (yr): 15.3 | 51% |
| Soni et al., 2017 [50] | India | C–S | 587 | University | [a]Mean age (yr): 16.2 (SD: 2.3) | 58% |
| Tamura et al., 2017 [53] | Japan | C–S | 295 | High school | Mean age (yr): 16.2 (SD: 0.9) | 57% |
| Tan et al., 2016 [41] | China | C–S | 1661 | High schools | Mean age (yr): 14.5 | 52% |
| Tao et al., 2017 [47] | China | C–S | 4747 | University | Mean age (yr): 19.2 (SD:1.4) | 42% |
| Thomée et al., 2012 [29] | Sweden | PC | 4163 | Swedish tax agency registry | Age (yr): 20-24 | 35% |
| Vernon et al., 2018 [70] | Australia | PC | 1101 | High schools | Mean age (yr): 13.5 | 43% |
| Visnijic et al., 2018 [48] | Italy | C–S | 785 | Universities | Mean age (yr): 20.8 (SD: 2.6) | 32% |
| Volungis et al., 2019 [49] | USA | C–S | 150 | College | Mean age (yr): 19.3 | 17% |
| Woods and Scott, 2016 [62] | Scotland | C–S | 467 | High school | Age (yr): 11-17 | – |
| Wu et al., 2015 [65] | China | C–S | 4747 | College | [b]Mean age (yr): 19.3 (SD: 1.4) | 71% |
| Ye et al., 2016 [24] | China | C–S | 2422 | Universities | Mean age (yr): 19.7 (SD: 1.2) | 59% |

a: High SM users.
b: males.
c: Nationally representative.
C-S: cross-sectional, M: male, PC: prospective cohort study, SD: standard deviation, SE: standard error, Yr: years, -: not reported.

Griffiths [63] specifically examined Facebook addiction and Wu et al. [65] only assessed general computer use including social media use. For mental health and sleep variables, symptoms of depression (n = 16/36) and sleep quality (n = 16/36) were commonly examined. Social media use, sleep, and mental health variables were either self-reported or assessed using questionnaires or validated scales.

*Risk of bias assessment*

Most of the studies had a low to moderate risk of bias (Fig. 2 and Fig. 3). The majority of cohort studies (n = 5/6) had a low risk of bias for selecting exposed and non-exposed subjects from the same population. Most studies showed a moderate risk of bias for the assessment of the exposure (n = 5/6) and outcome (n = 4/6). Three

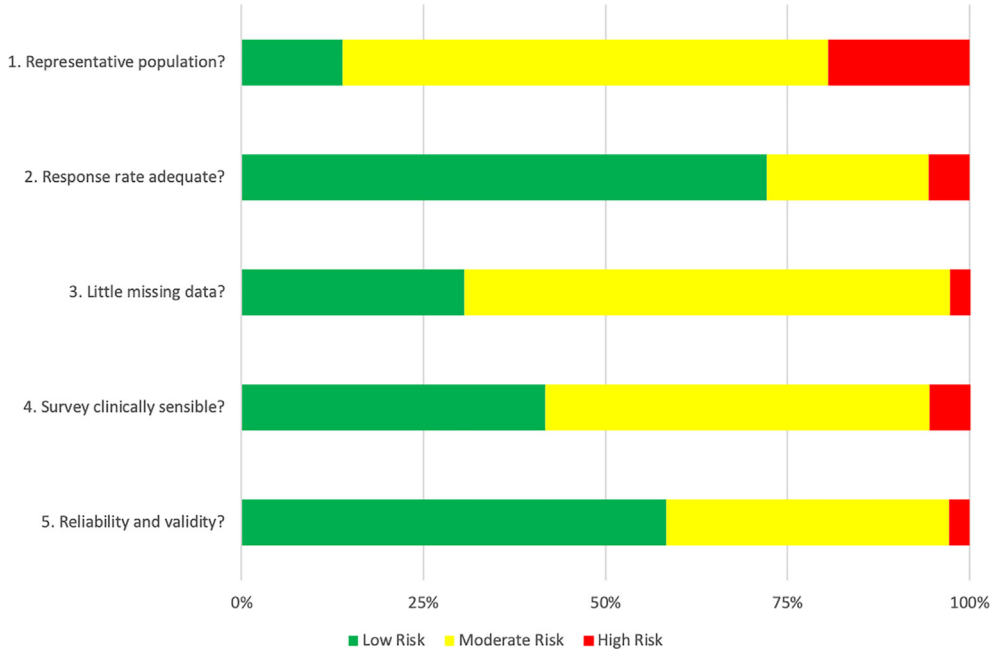



**Fig. 2.** Findings from the risk of bias assessment in thirty-six cross-sectional studies using the *CLARITY Risk of Bias Instrument for Cross-Sectional Surveys of Attitudes and Practices* [24].

of the six cohort studies had a high risk of bias regarding the presence of the outcome at the beginning of the study. The risk of bias was also low for the assessment of and adjustment for prognostic variables (n = 4/6), as well as matching exposed and unexposed variables associated with the outcome of interest (n = 4/6). Most of the studies were rated as having moderate risk of bias for providing adequate follow-up (n = 5/6), which also raises the possibility of selection bias [65].

For cross-sectional studies, most (n = 31/36) ranged from moderate to poor for obtaining a representative source sample. Some studies used convenience samples and most studies (n = 19/36) recruited their samples from academic institutions, thereby reducing the generalizability of the findings [66]. The cross-sectional studies rated well on adequate response rates from participants (n = 19/36) and on providing evidence for reliability and validity of the survey instruments (n = 17/36). Missing data was rated moderate for risk of bias for most studies as many did not report on this domain (n = 24/36). With regards to the clinical sensibility of the questionnaires used to assess social media exposure, fifteen studies of the 36 cross-sectional studies had a low risk of bias and thirteen of the 36 studies ranged from moderate to poor which increases the risk of information bias [66].

### *Synthesis of findings*

We synthesized the findings of the longitudinal and cross-sectional studies separately, and assessed findings pertaining to relationships reported between social media use and mental health, social media use and sleep quality, and sleep quality and mental health in youth populations ("See Table 2").

### *Social media and mental health*

Across all cohort studies, four found evidence suggesting that high frequency of social media use was a risk factor for poor mental health outcomes at follow up. Liu et al. [68] found long-time mobile phone use to be a significant risk factor for depression and anxiety in a sample of college students from China. Another study [30], using a sample of high-school students in China, reported that the risk of depression was significantly higher among severe and moderate internet users compared to normal internet users. Using longitudinal growth models, Vernon et al. [69] observed increases in mobile phone use associated with increases in depression, with sleep mediating the association in youth living in Australia. Hokby et al. [31] found that spending time on the internet indirectly predicted changes in poor mental health by predicting changes in sleep loss.

Thirty-three cross-sectional studies observed significant relationships between social media use and poor mental health in youth. Positive associations were observed in studies assessing excessive internet use (n = 15), frequent mobile phone use (n = 13), and general social media use (n = 4), with one study focusing on Facebook addiction [62]. Two studies [50,51] reported on sex differences in this relationship, and another two [30,54] reported on gender differences. Mixed findings were observed overall.

Sleep quality was reported as a mediator for associations between social media use and depressive symptoms across six cross-sectional studies. Of these studies, three were conducted using a sample of secondary students in China [32,39] and in Switzerland [58], one study included a sample of undergraduate students in Nepal [36], another used a convenience sample in the US [44], and one used a nationally representative sample in South Korea [59].

Interaction effects were assessed by two cross-sectional studies [35,46]. Park and colleagues [35] examined the moderating effects of depression on the relationship between internet use and sleep quality and observed that a decrease in sleep problems were associated with an increase in internet use among depressed students. Tao et al. [46] assessed the interaction effects of social media use and sleep on mental health outcomes in adolescents and found that sleep moderated the association between mobile phone use and negative mental health outcomes [46].

### *Social media use and sleep quality*

Five cohort studies found excessive social media use at baseline as a risk factor for poor sleep quality at follow up. Two of these

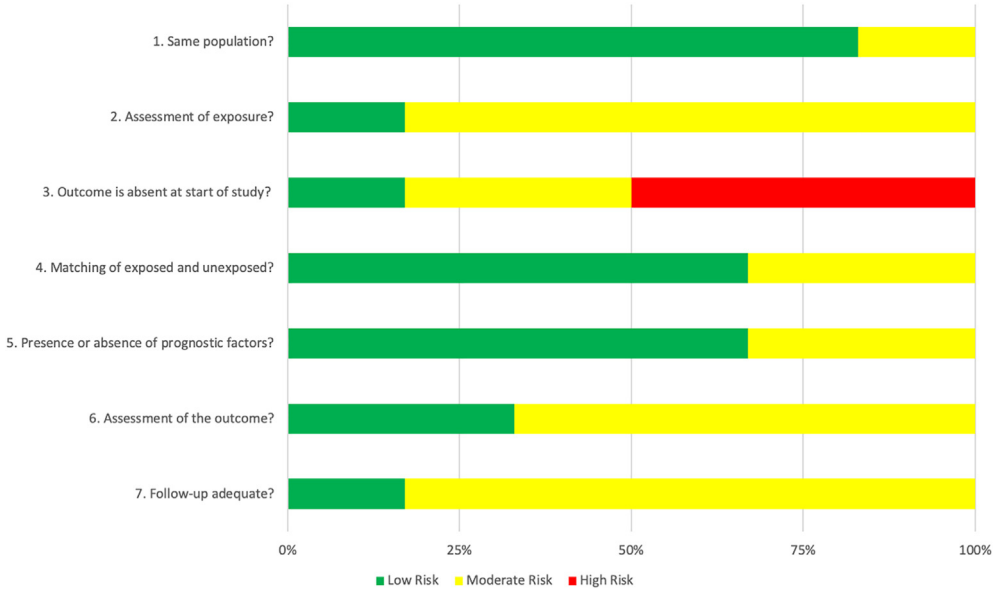



**Fig. 3.** Findings from the risk of bias assessment in six prospective studies using the *CLARITY Tool to Assess Risk of Bias in Cohort Studies* [25].

studies reported social media use as a risk factor for sleep disturbances: Thomee et al. [29] observed an association between excessive computer use and sleep disturbances in men, and Hokby et al. [28] reported that time spent on the internet was associated with sleep loss at follow-up among students. Both Rod et al. [67] and Liu et al. [69] observed mobile phone use as a risk factor for poor sleep outcomes, with Liu et al. [69] reporting negative and bidirectional associations between mobile phone use and weekday sleep duration and positive and bidirectional associations between mobile phone use and weekend sleep duration and compensation.

Significant associations between social media use and sleep quality were observed in twenty-four cross-sectional studies. Of these studies, twenty-three found positive associations between frequent social media use and poor sleep quality, with three studies using nationally representative samples of youth from Europe [30,58] and Asia [58] and another three studies using a convenience sample in China [39], US [44], and Bangladesh [63]. Among high-school students in Switzerland, Lemola et al. [59] found that night-time social media use was negatively associated with sleep duration and positively associated with sleep difficulties.

### *Sleep and mental health*

Significant findings pertaining to poor sleep quality and poor mental health outcomes, were observed in three cohort studies. Hokby et al. [28] and Thomee et al. [29] both reported significant associations between sleep loss due to social media use and mental health outcomes at follow up. Vernon et al. [70] found positive associations between increases in poor sleep behavior and depressed mood.

Sixteen cross-sectional studies found significant associations between poor sleep quality and poor mental health among young adults, with fifteen studies observing positive associations. Brunborg et al. [63] reported negative associations between poor mental health and weekday sleep duration. The study by Bhandari et al. [37] found an indirect effect between sleep quality and depressive symptoms, with social media use mediating the relationship.

## Discussion

This systematic review synthesized prior evidence on the interplay between social media use, sleep quality, and mental health in youth. In the 42 included studies, excessive internet use and mobile phone use were the most commonly examined forms of social media, with depression and anxiety being the most commonly assessed mental health outcomes. The findings of this review suggest an association between excessive social media use and poor mental health – specifically anxiety, depression, and psychological distress – among youth. However, it can be hard to discern whether spending extended periods of time on social media or frequently using such sites leads to poor mental health outcomes, or whether the use of social media might be a maladaptive coping mechanism for pre-existing mental health issues [45,69].

Regular checking of social media can contribute to anxiety related to “fear of missing out” (FOMO), which in turn can further aggravate compulsions to check social media sites [70,71]. The current COVID-19 lockdown led to prolonged school closures, resulting in youth spending more time at home and on social media platforms. As youth are separated from their peers for an extended period of time, they may experience changes in mental health, particularly an elevated amount of FOMO as a product of increased social media use [72].

Two included studies investigating social media use relative to bedtime [61,62] found that excessive use before and after bedtime were associated with higher levels of depressive symptoms. This is consistent with suggestions from Bhat and colleagues [73] that longer and more frequent checking of social media before bedtime, can increase general cognitive arousal based on the nature of news and discussions. Such habitual use of social media can also increase feelings of anxiety and distress [73].

Several of the cross-sectional and longitudinal studies in this review suggest sleep as a potential mediator of relationships between social media use and mental health (“see Table 2”). This finding highlights the potential mechanism of association for the three variables in terms of overall adolescent health. From a physiological perspective, extended use of social media at bedtime delays sleep onset and reduces sleep duration. Habitually

**Table 2**
Summary of findings.

| Author | Sleep measures | Mental health measures | Social media use measures | Main Findings |
|---|---|---|---|---|
| Adams and Kisler, 2013 [45] | Sleep quality[d] (Pittsburgh sleep quality index, PSQI), sleep patterns[b] (sleep diaries) | Depressive[b] symptoms (Beck depression index, BDI), anxiety (Adult manifest anxiety scale) | Technology use: number of times awakened to answer calls/texts,[a] time awake after each occurrence[a] (sleep diaries) | Associations between frequent technology with depressive and anxiety symptoms (sleep quality may mediate relationship), total awake time with poor sleep quality and depression and anxiety with poor sleep quality |
| Belanger et al., 2011 [31] | Sleep quantity[b] (self-reported) | Presence of depression[b] (Depressive tendencies scale) | Intensity of Internet use[a] (self-reported) | Females: High internet users associated with higher depressive scores and poor sleep<br>Males: High internet users associated with higher depressive scores |
| Bhandari et al., 2017 [37] | Sleep quality[a,d](PSQI) | Depressive symptoms[b] (Patient health questionnaire-9, PHQ-9) | Internet use[a,d] (Young's internet addiction test, YIAT) | Associations between high internet use with depressive symptoms (30.9% mediated by sleep quality) and poor sleep quality with depressive symptoms (16.5% mediated by internet addiction) |
| Brunborg et al., 2011 [63] | Sleep habits[b] (self-reported sleep time and duration), insomnia (Bergen insomnia scale) | Symptoms of anxiety[a] and depression[a] (Hospital anxiety and depression Scales, HADS) | Media use[a] (self-reported mobile and other device use frequency) | Positive association between anxiety and media use. Negative association between anxiety and depression with sleep habits |
| Chen et al., 2017 [55] | Sleep quality[a,b](PSQI) | Anxiety symptoms[a,b] (Zung self-rating anxiety scale, SAS[f]), depressive symptoms[a,b] (Center of epidemiological studies −depression scale, CES-D) | Smartphone addiction[a,b] (Smartphone addiction scale, SAS−short version) | Females: Smartphone addiction associated with anxiety, depression and poor sleep quality<br>Males: Smartphone addiction associated with anxiety and poor sleep quality |
| Cheung and Wong, 2011 [33] | Insomnia[a,d] (PSQI) | Depression[b] (General health questionnaire, GHQ-12) | Internet addiction[a] (Chinese internet addiction scale) | Associations between internet use with depression (sleep quality may mediate relationship) and sleep quality and insomnia with depression. |
| Demirci et al., 2015 [44] | Sleep quality[a,b](PSQI) | Depression[a,b] (BDI), anxiety[a,b] (Beck anxiety inventory) | Smartphone use[a] (Smartphone addiction scale, SAS[g]) | Associations between high smartphone use with depressive and anxiety symptoms and with poor sleep quality |
| Dewi et al., 2018 [56] | Sleep disturbance[b] (Insomnia severity index, ISI) | Depressive symptoms[b] (Kutcher adolescent depression scale) | Duration of smartphone use at night[a] (self-reported) | Positive association between smartphone use with depressive symptoms and sleep problems |
| Do et al., 2013 [34] | Sleep duration on weekday[a] (self-reported) | Depressive symptoms[b] (self-reported) | Time (in minutes) spent on internet[a] (self-reported) | Association between shorter sleep duration with increased depressive symptoms among high internet user |
| Eyvazlou et al., 2016 [57] | Sleep quality[a] (PSQI) | Anxiety,[a,b] severe depression[a,b] (GHQ-12) | Overuse of mobile phones[a,b] (Cell phone overuse scale) | Associations between high mobile phone use with greater sleep problems and poor sleep quality with depression and anxiety |
| Hokby et al., 2016 [28] | Sleep loss[a,c] (self-reported) | Depression,[b] stress,[b] anxiety[b] (Depression anxiety stress scale-42, DASS-42) | Internet use and behavior (self-reported) | Associations between internet use with increased sleep loss and sleep loss from internet use with changes in mental health over time |
| Ikeda and Nakamura, 2014 [52] | Hours sleeping[c] (self-reported) | Depressed[b] and anxious mood[b] (Mood inventory) | Hours of mobile phone use[a] (self-reported) | Associations between high mobile phone use with increased depression after adjusting for covariates (males and females) and high mobile phone use with increased depressed mood and anxious mood (among males but not females) |
| Islam and Hossin, 2016 [42] | Sleep duration[a] (self-reported) | Psychological distress[b] (GHQ-12) | Problematic internet use[a] (YIAT) | Association between high problematic internet use with psychological distress |
| Kadam et al., 2016 [58] | Sleep quality[b] (PSQI) | Depressed feelings[a] (self-reported) | Mobile messaging applications measured as lifestyle factors[a] (self-reported) | Associations between sleep disturbances with higher frequency of mobile messaging and increased feelings of depression |
| Kitazawa et al., 2018 [35] | Sleep quality[a] (PSQI) | Depressive Symptoms[a] (CES-D), anxiety (Trait subscale of state-trait anxiety inventory) | Internet use[b] (YIAT) | Associations between depressive state, anxiety tendencies and poor sleep quality with problematic internet use |
| Kojima et al., 2019 [39] | Degree of sleepiness upon awakening,[c] bedtime[c] (self-reported) | Depressive symptoms[a] (Birleson depression self-rating scale) | Internet use[b,c] (YIAT) | Positive associations between depressive symptoms, late bedtime and daytime sleepiness are positively associated with problematic internet use |
| Kootesh et al., 2016 [38] | Sleep quality[a,b] (PSQI) | Anxiety[a,b] (Mental health checklist) | Internet addiction[a,b] (YIAT) | High internet use is positively associated with anxiety and poor sleep quality |
| Lam and Peng, 2010 [30] | | | Internet use[b] (YIAT) | |

(*continued on next page*)

**Table 2** (*continued*)

| Author | Sleep measures | Mental health measures | Social media use measures | Main Findings |
|---|---|---|---|---|
| | Sleeping hours[c] (self-reported) | Anxiety[a] (SAS[f]), depression[a] (Zung Self-rating Depression Scale, SDS) | | Positive association between pathological internet use and depression. No relationship between internet use and anxiety |
| Lange et al., 2017 [59] | Insomnia[b] (self-reported) | Anxiety,[c] depressive symptoms[c] (Emotional problems subscale of strengths and difficulties questionnaire) | Mobile phone and computer/ internet use[a] (self-reported) | Increased computer/internet use associated with insomnia complaints among males |
| Lemola et al., 2014 [60] | Sleep duration[d] (self-reported), sleep difficulties (ISI) | Depressive symptoms[b] (CES-D-German) | Electronic media use[a] (before bedtime), general daily duration of electronic media use[a] (self-reported) | Associations between media use with depressive symptoms (sleep may mediate relationship) and sleep difficulties |
| Li et al., 2017 [40] | Insomnia[d] (PSQI) | Depression[b] (CES-D) | Internet addiction[a] (Young's diagnostic questionnaire), social networking addiction[a] (Online social networking addiction scale) | Associations between internet addiction and online social networking with depression (insomnia may mediate this relationship) and insomnia with depression |
| Liu et al., 2019 [69] | Sleep quality[b] (PSQI), weekday/weekend sleep patterns (self-reported) | Depressive symptoms[b] (BDI), anxiety (SAS[f]) | Duration of mobile phone use per day[a] (self-reported) | Long-time mobile phone use at baseline is associated with depressive and anxiety symptoms at follow-up. Long-time mobile phone use is negatively and bidirectionally associated with weekday sleep duration; positively and bidirectionally associated with weekend sleep duration and compensation |
| Mamun and Griffiths, 2019 [64] | Sleep duration per day[a] (self- reported) | Depressive symptoms[a] (PHQ-9) | Facebook addiction risk[b] (Bergen facebook addiction scale), amount of time spent on Facebook (self-reported) | Associations between increased depressive symptoms and sleep disturbances with Facebook Addiction |
| Mamun et al., 2019 [43] | Sleep duration[c] (self-reported) | Depression,[a] anxiety,[a] stress[a] (DASS-21) | Problematic internet use[b] (YIAT) | Associations between problematic internet use with depression, anxiety and stress. No relationship between sleep duration and internet use |
| Nishida et al., 2019 [51] | Hours sleeping[c] (self-reported) | Depression[b] (CES-D) | Hours of smartphone use[a] (self-reported) | Associations between high smartphone use with depression among females but not males. No relationship between smartphone use and sleep |
| Park et al., 2018 [36] | Sleep problems[b] (ISI), sleep-wake patterns (School sleep habits survey, School-SHS) | Depressive symptoms[e] (Children's depression inventory) | Internet use[a] (YIAT) | Depressed group: decrease in sleep problems are associated with increased internet use (depression may moderate this relationship)<br>Non-depressed group: increase in sleep problems are associated with increase in internet use |
| Peltzer and Pengpid, 2015 [32] | Sleep duration[a] (self-reported) | Depressive[b] symptoms (CES-D) | Time of internet use[a] (self-reported) | Associations between high internet use with severe depressive symptoms and irregular sleep duration with depressive symptoms |
| Rod et al., 2018 [67] | Sleep duration,[b] sleeping problems[b] tiredness[b] (self-reported), sleep disturbances[b] (Karolinska sleep questionnaire) | Anxiety,[b] depressive state[b] (self-reported), depressive symptoms[b] (Major depression inventory) | Smartphone activity during sleep[a] (smartphone use data), smartphone-interrupted sleep[a] (self-reported) | Associations between more frequent smartphone-interrupted sleep with shorter sleep duration. No differences in mental health symptoms between those with and without smartphone-interrupted sleep |
| Rosen et al., 2016 [46] | Sleep problems[b] (Medical problems study sleep measure) | Anxiety,[a] dependence related to being without phone[a] (Media and technology usage and attitudes scale) | Daily smartphone usage[a] (subscale of media and technology usage and attitudes scale) | Associations between higher anxiety/ dependence with higher daily smartphone usage and increased sleep problems. |
| Seo et al., 2017 [61] | Sleep habits,[d] duration, weekday and weekend bedtime[d] (self-reported), sleep disturbances[d] (Global sleep assessment questionnaire) | Depression[b] (BDI) | Media use-related variables, use of media on weekdays and before sleep,[a] types of media used[a] (self-reported) | Associations between later media use with increased depression (sleep may mediate this relationship) and increased sleep duration and increased sleep duration with decreased depression |
| Soni et al., 2017 [50] | Sleep quality[a,b] (PSQI) | Psychological distress[a,b] (GHQ-12), depression,[a,b] anxiety,[a,b] stress[a,b] (DASS-21) | Smartphone use[a,b] (SAS[g]) | Associations between increased smartphone use with increased depression, anxiety and poor sleep quality |
| Tamura et al., 2017 [53] | Insomnia[b] (Athens Insomnia Scale) | Depression[b] (CES-D) | Mobile phone use[a] (self-reported) | Associations between increased mobile phone use with social networking sites, online chat and insomnia |

**Table 2** (*continued*)

| Author | Sleep measures | Mental health measures | Social media use measures | Main Findings |
|---|---|---|---|---|
| Tan et al., 2016 [41] | Sleep quality[b] (PSQI) | Depressive symptoms[a,d] (CES-D) | Problematic internet use[a,d] (YIAT), duration and impact of internet use (self-reported) | Associations between problematic internet use with increased depressive symptoms and sleep disturbance (depression may mediate this relationship) and depressive symptoms with sleep disturbance (internet use may mediate this relationship) |
| Tao et al., 2017 [47] | Sleep quality[a,e] (PSQI) | Anxiety symptoms[b] (SAS[f]), depressive symptoms[b] (CES-D) | Mobile phone use[a,e] (Self-rating questionnaire for adolescent problematic mobile phone use) | Associations between problematic phone use with anxiety and depressive symptoms (sleep may moderate this relationship) and poor sleep quality. |
| Thomée et al., 2012 [29] | Sleep disturbances[b] (self-reported difficulties falling asleep, fragmented sleep, premature awakening) – adapted from Karolinska sleep questionnaire | Depressive symptoms[b] and current stress[b] (adapted from Prime-MD) | Computer[a] and mobile phone use[a] (self-reported) | Females: computer use is positively associated with current stress, depressive symptoms, and sleep disturbances. Computer use causing sleep lost is associated with sleep disturbances, current stress and depression<br>Males: computer use is positively associated with sleep disturbances. Computer use causing sleep lost associated with current stress |
| Vernon et al., 2018 [70] | Sleep behavior[d] (SSHS) | Depressed mood[b] (Questionnaire from Michigan life transitions) | Mobile phone use at night-time[a] (self-reported) | No association between changes in night-time mobile phone use with changes in depressed mood.<br>Associations between higher mobile phone use at baseline is associated with poorer sleep behavior at follow up and poor sleep behavior with depressed mood |
| Visnijic et al., 2018 [48] | Sleep duration[a] (self-reported) | Depression,[b] anxiety,[b] stress[b] (DASS-42) | Manner and intensity of mobile phone use[a] (self-reported) | Associations between increased texting and less internet browsing with anxiety and depression and keeping the phone < 1 m away with increased stress and later bedtime with more internet browsing and texting |
| Volungis et al., 2019 [49] | Sleep quality[a] (PSQI) | Social Emotional Distress[a,b] (Outcome questionnaire-45.2; symptom distress subscale emphasizes depression and anxiety) | Smartphone use[a] (SAS[g]) | Associations between high smartphone use with increased depression, anxiety and poor sleep quality |
| Woods and Scott, 2016 [62] | Sleep quality[b] (PSQI) | Anxiety[a] and depressive[a] symptoms (HADS) | Overall and night-time social media use[a] (self-reported) | Associations between high anxiety and depressive symptoms with increased social media use and poor sleep quality and night-time media use with poor sleep quality |
| Wu et al., 2015 [65] | Sleep quality[b] (self-reported) | Anxiety[b] (SAS) and Depression[b] (CES-D) | Hours of computer screen time[a] (self-reported) | Associations between high screen time with increased depressive and anxiety symptoms and poor sleep quality |
| Ye et al., 2016 [24] | Sleep quality[a] (PSQI) | Depression[b] (SDS), anxiety[b] (SAS[f]) | Internet addiction[a] (YIAT) | Associations between internet addiction with anxiety and depression and sleep disturbance with anxiety and depression |

Analysis of variables legend.
a = Exposure.
b = Outcome.
c = Confounder.
d = Mediator.
e = Moderator.
Abbreviations and acronyms:
BDI: Beck depression inventory, CES-D: Center of epidemiological studies-depression scale, DASS-42 (-21): Depression anxiety stress scale - 42 item (- 21 item), GHQ-12: General health questionnaire-12 item, HADS: Hospital anxiety and depression scale, ISI: Insomnia severity index, PHQ-9: Patient health questionnaire-9 item, PSQI: Pittsburgh sleep quality index, SASf: Zung self-rating anxiety scale, SASg: Smartphone addiction scale, SDS: Zung self-rating depression scale, SHS: Sleep habits survey (School), YIAT: Young's internet addiction test.

delayed sleep onset can result in a shift in circadian rhythm, leading to persistent sleep difficulties and reduced mood [45,74]. Exposure to electromagnetic fields from mobile phones – a common portal for social media – may lead to reduced melatonin output, consequently delaying sleep onset and thereby increasing the likelihood of psychiatric symptoms [75–77]. Social media use around bedtime can delay sleep onset and reduce sleep duration, especially due to the constraints of waking early for school or work. When combined with the cognitive and often emotional arousal that arises with social media content appraisal, such reduced sleep can be restless and lead to negative affect when performed habitually [45,73,77,78].

Sleep disturbances and poor sleep quality may arise from the use of interactive technological activities (particularly social media)

due to increased mental stimulation and temporal displacement, compared to more passive activities such as watching television [69]. International studies of youth from various regions have reported adverse effects of social media use on sleep, such as reduced sleep duration, poor sleep quality, and more frequent disturbances [14,24,36–38,44,67,75,79]. In turn, such negative impacts on sleep can have adverse health effects, particularly reduced psychological well-being and cognitive functioning [80–82]. These health impacts are pertinent to youth, who do not always have the ability to compensate for the lack of sleep by simply sleeping longer [79].

Most studies were conducted in Asia (n = 27), followed by Europe (n = 9) and North America (n = 3). One study was conducted using samples from multiple continents [32], while another study used an Australian sample [70]. As such, the observed associations between excess social media use, sleep, and mental health symptoms may be explained to some extent by cultural differences surrounding sleeping habits and perceptions of mental health [6,83–85]. However, inconsistencies in measurement tools and analyses used in the included studies make these cultural differences difficult to confirm.

Our review highlights several gaps in the current evidence base on the interrelationships between social media use, sleep, and mental health. First, most of the included studies used a cross-sectional design, making it difficult to assess temporality in the observed associations.

Only six studies used a longitudinal design, of which only one [29] could ascertain that the mental health conditions were not present at baseline. Prior studies were also limited in regard to the representativeness of the study samples, as most were convenience samples recruited from a post-secondary setting. Additionally, all included studies used self-reported measures of sleep duration and quality, as well as social media and technology use. Future studies would benefit from the use of more objective methods of measuring sleep quality and technology use (actigraphy, screen time monitors on smartphones) as well as adopting longitudinal study designs to clearly establish temporality of the observed associations. Furthermore, only six studies examined gender differences in social media use, in relation to sleep quality and mental health, and the findings were equivocal. As such, future studies could explore gender differences in social media use to help clarify associations with sleep and mental health in youth populations.

### *Limitations*

Our systematic review had several limitations. First, only one reviewer extracted the data independently. However, as there was a second reviewer available to verify the data, this may have helped mitigate any potential errors or bias that could have occurred during the data extraction process. Another limitation was that the terms “youth”, “young adults” and “adolescents” were used interchangeably across all studies. In our review, we defined “youth” as individuals aged 16–25 years old and expanded our inclusion criteria to include groups aged 12–30 years old. Many of the studies simplified the term “youth” to include “adolescent” and “young adult” populations, thereby including an inconsistent categorization of youth which varied by different age ranges. Unless an age range was explicitly provided, it was difficult to ensure that the findings from the studies were consistent with our prespecified definition of youth. We also placed greater focus on youth populations, thereby limiting the generalizability of our findings. Although youth are commonly associated with social media use, given the widespread nature of social media, broadening our search to additional populations, such as older age groups, could help increase the generalizability of our results. Finally, there exists much heterogeneity in measures used in the included studies, precluding a meta-analysis [27].

## Conclusions

The current review examines the interplay between social media use, sleep quality and duration, and poor mental health. Despite heterogeneity in the included studies, longitudinal studies suggest poor sleep quality and frequent sleep disturbances may partially explain the association between excessive social media use and poor mental health outcomes. Cross-sectional studies, inherent in their design, cite more equivocal and multidirectional effects between the three health variables. Social media use can contribute directly to poorer mental health outcomes, or indirectly through poorer sleep. The directionality and strength of such associations should be further elucidated using prospective cohort studies with representative youth samples. These findings may have public health implications given the widespread use of social media, the high prevalence of sleep problems, and increasing burden of mental health disorders in current societies.

**Practice points**

Awareness of the potential relationships between social media use, sleep quality and mental health may be useful to:

1. Improve sleep quality and other sleep outcomes in youth.
2. Improve mental health outcomes such as anxiety, depression and psychological distress.
3. Promote self-regulated use of social media applications.

**Research agenda**

Social media use warrants future studies to:

1. Clarify the strength and directionality of relationships between social media use, sleep quality and mental health using a prospective study designs with representative youth samples.
2. Use objective and consistent measures to quantify sleep quality and social media use.
3. Examine gender differences in social media use.
4. Assess these relationships in other technologically literate populations.

## Conflicts of interest

The authors do not have any conflicts of interest to disclose.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.smrv.2020.101414.

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